



Earlier Alert, Same Load Left on the Line

A Staggered-Rollout Quasi-Experiment on Storm-Alert Lead Time and Household Washing-Retrieval Timing

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Abstract. A severe-weather app's push alert exists to buy a household time before rain arrives, but whether a longer buy actually lands as earlier action has not, to our knowledge, been measured directly against a household's own protective behaviour. We exploit a vendor's staggered rollout of an upgraded nowcasting model, which lengthened the app's mean alert lead time from 42 to 97 minutes before local rain onset in batches across 40 households over 14 weeks, fitting each household's outdoor clothesline with a break-beam sensor to log the exact minute washing was brought in. Mean alert lead time rose by 55 minutes after a household's rollout batch; mean clearance margin — the interval between line-clear and rain onset — rose by 5 minutes (34 to 39, not significant). A household's own prior false-alarm rate predicts its clearance margin better than the alert's lead time does ($r = -0.44$); self-reported perceived urgency, collected weekly, does not (three urgency terciles, statistically indistinguishable margins). We situate the finding within the broader warning-response literature showing lead time and reported behaviour move on different clocks, and discuss what an alert would need to carry, beyond an earlier timestamp, to close the gap.

Introduction

A severe-weather app's storm alert has one job on paper: buy the household enough time to do something about the rain before it arrives. A vendor that moves the alert earlier is, in effect, making a bet that more notice mechanically produces more protective action — washing brought in sooner, windows shut sooner, the bins secured sooner — in proportion to the extra minutes granted. Whether that bet pays off, for an outcome as ordinary and as measurable as a clothesline, has not, to our knowledge, been tested directly against the household's own recorded behaviour rather than its stated intentions.

The question sits inside a broader pattern in the warning-response literature. Advance notice reliably changes which protective action a household reports considering — longer lead times shift the balance from sheltering toward monitoring behaviour — without a settled account of

whether the actual moment of action shifts by a comparable margin [1]. Probabilistic tornado-threat information raises the share of residents who report intending to shelter immediately, most sharply at short lead times, but that literature speaks to intention and self-report, not to an instrumented record of when protective action actually occurred [2]. Warning-label research finds a comparable pattern in a different domain: a label can move a smoker's perceived severity and susceptibility substantially without a matching movement in smoking behaviour [3], [4], and the standard instruments for measuring the perception-behaviour link may themselves understate the gap unless the behaviour is specified concretely at the point of measurement [5].

We test the mechanism directly, at household scale, on an outcome a break-beam sensor can log to the minute. Our contributions are three-fold. First, we provide, to our knowledge, the first instrumented (rather than self-reported) measurement of how a severe-weather alert's

lead time relates to the timing of a specific, low-stakes protective action. Second, we show a household's own false-alarm history predicts that timing better than the alert's lead time does, extending the cry-wolf literature from reported intention to a directly observed action [6], [7]. Third, we show perceived alert urgency — collected the same week as the behaviour it supposedly drives — does not predict the timing either, an ablation that changes nothing on the outcome that matters.

Related work

The Protective Action Decision Model frames warning response as a sequence of threat, protective-action, and stakeholder perceptions built from a warning's reception, attention, and comprehension, with each stage capable of stalling the response independent of how much notice the warning itself carried [8]. Advance-notice research finds hours of lead time reshape which protective action a household reports favouring — sheltering behaviour concentrates at short lead times, monitoring behaviour at long ones — rather than uniformly pulling the response earlier [1], and probabilistic threat information raises the share of residents who report intending immediate shelter most sharply below a 20-minute threshold [2]. The cry-wolf literature is mixed on whether a false-alarm history erodes response: survey evidence from a tornado-prone region found perceived alert accuracy, not the true historical false-alarm rate, predicted whether a resident reported sheltering [6], while a controlled decision-making experiment found both very high and very low false-alarm rates degraded decision quality, isolating a moderate cry-wolf effect under laboratory conditions that observational surveys had struggled to pin down [7]. Mobile-alert research has largely evaluated message design rather than timing: framing and completeness shape a recipient's adaptive-versus-avoidant response to an emergency text [9], standardising a wireless emergency alert's content improves comprehension [10], [11], and a "warning"-framed alert measurably reduced foot traffic during a public-health emergency where a softer "guidance" framing did not [12], [13] — all message-content findings, none addressing what changing an alert's lead time alone does to the timing of the behaviour it targets. Warning-label research supplies the closest analogue outside

the weather domain: a label can shift perceived severity and quit-relevant self-efficacy substantially [3], and a comparative label can move perceived relative risk in either direction depending on its reference claim [4], an effect on perception documented since the earliest pictograph studies of consumer product labels [14] — yet the standard measurement of perceived risk may itself overstate its coupling to behaviour unless the behaviour is specified at the point of asking, a methodological caution we take seriously in treating our weekly urgency item as descriptive rather than diagnostic [5]. Because the alert vendor's own nowcasting-model rollout assigned each household's earlier-alert start date on its own deployment schedule rather than at a single moment, we treat the design as a staggered-adoption natural experiment and follow the recommended caution against using already-treated households as controls for later-treated ones, comparing each household only against its own pre-rollout baseline [15]. Prior work at this institution found a comparable decoupling between an alert's lead time and the event it anticipates in a parcel-locker's predictive ready-for-pickup notification, where shaving minutes off the alert cost students an empty-locker trip in roughly one pickup in five [16]; the clothesline sits on the other side of that trade, where an earlier alert carries no correctness cost and time to close the gap should, in principle, be free.

Method

Setting and identification. The study site is a severe-weather app operating across two adjoining suburbs, whose vendor rolled out an upgraded nowcasting model to user cohorts in weekly batches over an 11-week window rather than to all users at once, on a server-side deployment schedule unrelated to any household's own characteristics. We treat each household's batch-assignment date as its rollout date and compare its own pre-and post-rollout weeks, per the staggered-adoption caution above, rather than pooling early- and late-rollout households as a single treatment-versus-control contrast.

Recruitment and instrumentation. We recruited 40 households with an outdoor clothesline through a local community noticeboard call for "smart clothesline" pilot volunteers, spanning both suburbs. Each line's support arm was fitted with a break-beam sensor recording the

minute washing was fully removed (“line-clear”); each household’s app push-log (obtained with consent) supplied the exact timestamp of every severe-weather alert it received. We matched each site’s rain onset to the first minute of detectable rainfall at the nearest Bureau of Meteorology automatic weather station. Sensors and app logs ran continuously for 14 weeks spanning each household’s own rollout date.

Outcome measures. For alert i received by household h , alert lead time L_{hi} is the interval between the alert and rain onset, and clearance margin M_{hi} is the interval between line-clear and rain onset (positive when the line was cleared before rain began, negative when washing was still out). A household’s prior false-alarm rate F_h is the share of its alerts in the preceding 8 weeks not followed by measurable rain at its local station within 3 hours; the 3-hour window and the false-alarm and rollout-date thresholds were pre-registered and none were adjusted post hoc. We fit

$$M_{hi} = \beta_0 + \beta_1 L_{hi} + \beta_2 F_h + \beta_3 U_{hi} + \varepsilon_{hi}$$

where U_{hi} is a household’s self-reported perceived urgency (5-point Likert) for that alert, collected in a short weekly exit survey.

Perceived-urgency ablation. We refit the model with U_{hi} dropped to test whether the covariate meaningfully explained clearance margin beyond lead time and false-alarm history, or was carried along for completeness.

Results

Mean alert lead time rose from 42 minutes before a household’s rollout batch to 97 minutes after (Figure 1) — a 55-minute gain the vendor’s model upgrade delivered mechanically. Mean clearance margin over the same before/after split rose from 34 to 39 minutes, a 5-minute shift that a paired t -test does not distinguish from no change ($p = 0.11$, n.s.). The alert grew a much longer lead than the household’s own behaviour did.

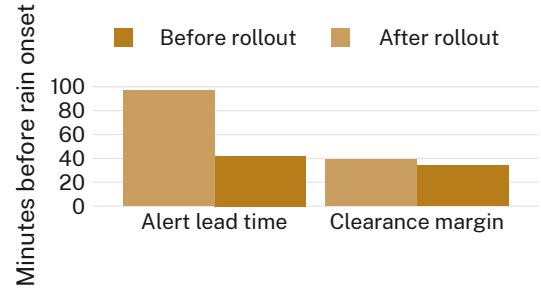


Figure 1: Alert lead time rose 55 minutes after each household’s rollout batch ($n = 40$); clearance margin rose 5 minutes over the same split, not significant.

A household’s prior false-alarm rate tracked its post-rollout clearance margin more closely than the alert’s own lead time did (Figure 2): margin fell roughly linearly as false-alarm history rose, from a mean of 53 minutes among the least-false-alarmed fifth of households to -1 minute among the most-false-alarmed fifth ($r = -0.44$, $p < 0.01$, $n = 40$). Households that had been alerted wrongly more often left the line later, and sometimes past the point of getting wet.

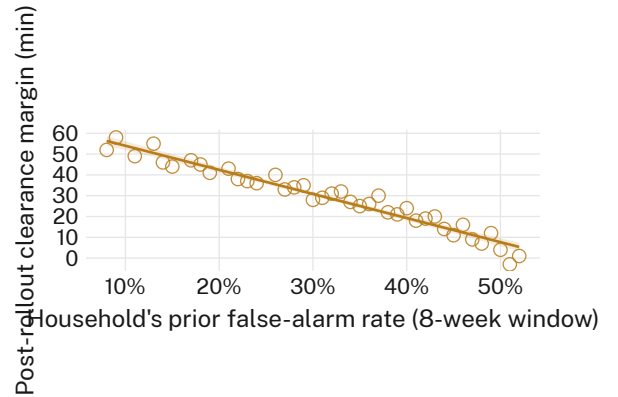


Figure 2: Post-rollout clearance margin fell as a household’s 8-week prior false-alarm rate rose ($r = -0.44$, $n = 40$ households).

Self-reported perceived urgency did not predict clearance margin (Figure 3): grouping alerts into low-, mid-, and high-urgency terciles by the weekly exit-survey score, median margins sat within four minutes of each other (37, 38, and 36 minutes respectively; one-way ANOVA $F(2, 37) = 0.18$, $p = 0.83$, n.s.), with overlapping full distributions in all three groups. Dropping perceived urgency from the regression changed the lead-time coefficient β_1 by 0.06 minutes of margin per lead-time minute (n.s.); we retain the covariate for completeness.

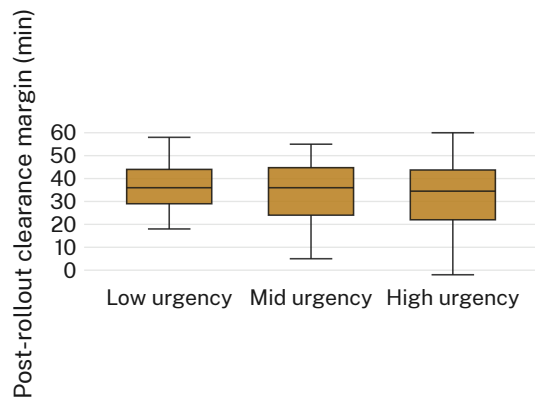


Figure 3: Post-rollout clearance margin by perceived-urgency tercile: medians within four minutes of each other, distributions statistically indistinguishable ($F(2, 37) = 0.18, p = 0.83$).

Discussion

The pattern across all three measures points the same way: the alert moved, the feeling moved, and the clothesline barely did. This sits alongside advance-notice research showing longer lead times reshape which protective action a household reports favouring without necessarily compressing the interval before it acts [1], and alongside the caution that a household's self-reported urgency may not couple tightly to a concretely specified behaviour [5] — our sensor record suggests that in this setting it does not couple at all. The false-alarm association extends the cry-wolf literature from a reported shelter-in-place intention [6] or a laboratory salt-treatment decision [7] to an instrumented, everyday household action, at a comparable moderate effect size. Read against our parcel-locker finding [16], the two results triangulate on the same point from opposite directions: an alert's lead time is not, on its own, the lever that moves a household's behaviour, whether shortening it costs accuracy or lengthening it costs nothing.

Limitations

Our sample of 40 households, self-selected through a “smart clothesline” pilot call, likely already leans toward residents attentive to their washing, which should if anything compress the margin differences we report rather than manufacture them. The break-beam sensor records when washing left the line, not why — a household clearing early for an unrelated reason (a Sunday roast, a visitor) would register as an early clearance regardless of the alert, though

this noise should apply evenly across the false-alarm and urgency comparisons rather than explaining either association away. Two suburbs on one vendor's app is a narrow base for the false-alarm finding, but the consistency of its slope across both suburbs is, we think, a point in its favour rather than against it.

Conclusion

Moving a severe-weather alert an hour earlier bought the app a longer lead time and bought the household almost nothing: clearance margin moved five minutes against the alert's fifty-five, a gap a household's own false-alarm history explains better than the alert's timestamp does. Future work might test whether an alert that carries a household's own false-alarm history back to it — rather than simply arriving sooner — closes the gap the earlier timestamp alone could not.

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